

Experiment 2: Java Basics

Prime, Palindrome and Command Line Arguments

1. Question

Write a Java program to:

- 1 Check whether a given number is a Prime number.
- 2 Check whether a given number is a Palindrome number.
- 3 Accept two numbers through Command Line Arguments and find their sum.

2. Steps

- 1 Start the program.
- 2 Create a Scanner object to read input from the keyboard.
- 3 Read a number to check whether it is prime.
- 4 Call the isPrime() method.
- 5 Display whether the number is prime or not.
- 6 Read another number to check whether it is a palindrome.
- 7 Call the isPalindrome() method.
- 8 Display whether the number is a palindrome or not.
- 9 Check whether two command-line arguments are provided.
- 10 Convert the command-line arguments from String to int.
- 11 Calculate and display their sum.
- 12 Close the Scanner.
- 13 Stop the program.

3. Algorithm

A. Prime Number

- 1 Start.
- 2 Read integer n.
- 3 If $n < 2$, return false.
- 4 Repeat from $i = 2$ to $n/2$.
- 5 If $n \% i == 0$, the number is not prime.
- 6 Otherwise, the number is prime.
- 7 Stop.

B. Palindrome Number

- 1 Start.
- 2 Read integer n.
- 3 Store the original number in original.
- 4 Set reverse = 0.
- 5 Extract the last digit using $n \% 10$.
- 6 Add the digit to reverse.
- 7 Remove the last digit using $n / 10$.
- 8 Repeat until n becomes 0.

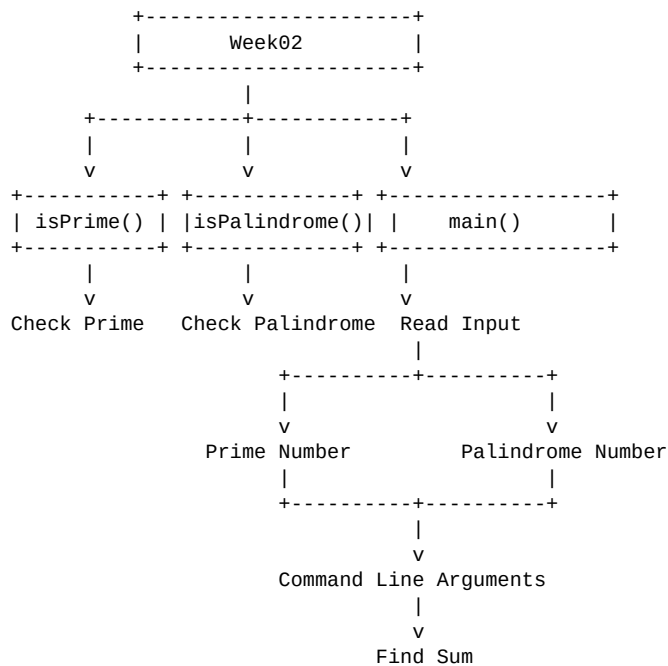
- 9 Compare original with reverse.
- 10 If both are equal, the number is a palindrome.
- 11 Otherwise, it is not a palindrome.
- 12 Stop.

C. Command Line Arguments

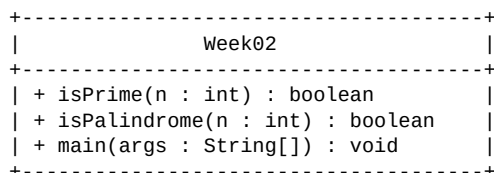
- 1 Start.
- 2 Check whether args.length >= 2.
- 3 Convert args[0] and args[1] into integers.
- 4 Add the two numbers.
- 5 Display the first number, second number and sum.
- 6 If two arguments are not provided, display No command line arguments provided.
- 7 Stop.

4. UML Diagram

Activity / Structure Diagram



UML Class Diagram



5. Source Code

```

import java.util.Scanner;

public class Week02 {

    // Method to check whether a number is Prime
    public static boolean isPrime(int n) {

```

```

        if (n < 2) {
            return false;
        }
        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                return false;
            }
        }
        return true;
    }

    // Method to check whether a number is a Palindrome
    public static boolean isPalindrome(int n) {
        int original = n;
        int reverse = 0;

        while (n != 0) {
            int digit = n % 10;
            reverse = (reverse * 10) + digit;
            n = n / 10;
        }
        return original == reverse;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // 1. Prime number check
        System.out.print("Enter a number to check Prime: ");
        int num1 = sc.nextInt();
        if (isPrime(num1)) {
            System.out.println(num1 + " is a Prime number.");
        } else {
            System.out.println(num1 + " is not a Prime number.");
        }

        System.out.println();

        // 2. Palindrome number check
        System.out.print("Enter a number to check Palindrome: ");
        int num2 = sc.nextInt();
        if (isPalindrome(num2)) {
            System.out.println(num2 + " is a Palindrome number.");
        } else {
            System.out.println(num2 + " is not a Palindrome number.");
        }

        System.out.println();

        // 3. Command Line Arguments
        System.out.println("--- Command Line Arguments ---");
        if (args.length >= 2) {
            int a = Integer.parseInt(args[0]);
            int b = Integer.parseInt(args[1]);
            int sum = a + b;

            System.out.println("First number : " + a);
            System.out.println("Second number : " + b);
            System.out.println("Sum : " + sum);
        } else {
            System.out.println("No command line arguments provided.");
        }

        sc.close();
    }
}

```

Compilation and Execution

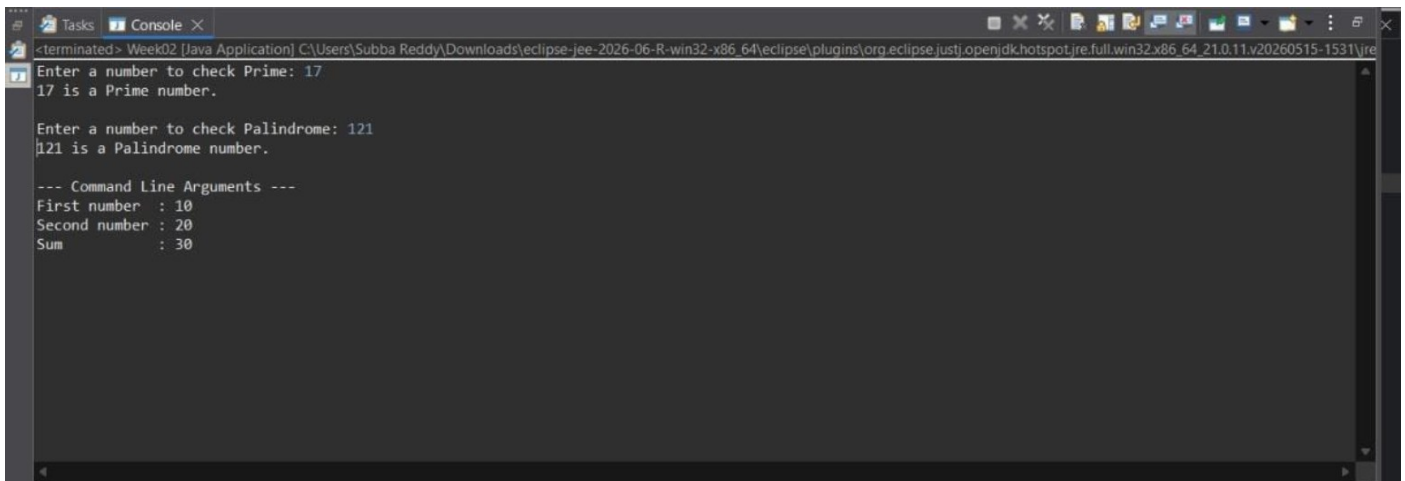
```
javac Week02.java  
java Week02 10 20
```

6. Output

```
Enter a number to check Prime: 17  
17 is a Prime number.  
  
Enter a number to check Palindrome: 121  
121 is a Palindrome number.  
  
--- Command Line Arguments ---  
First number   : 10  
Second number  : 20  
Sum            : 30
```

Result: The program was executed successfully and the outputs were verified.

Output Screenshot



```
<terminated> Week02 [Java Application] C:\Users\Subba Reddy\Downloads\eclipse-jee-2026-06-R-win32-x86_64\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre
Enter a number to check Prime: 17
17 is a Prime number.

Enter a number to check Palindrome: 121
121 is a Palindrome number.

--- Command Line Arguments ---
First number : 10
Second number : 20
Sum : 30
```